

AI-BASED FAKE REELS & NEWS DETECTOR

PROJECT REPORT

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology in Computer Science and Engineering

Academic Year: 2025-2026

CERTIFICATE

This is to certify that the project entitled “**AI-Based Fake Reels & News Detector**” is a bonafide work carried out for academic purposes under the Department of Computer Science and Engineering.

ACKNOWLEDGEMENT

We express our sincere gratitude to our faculty members, mentors, and friends for their valuable support during the development of this project. We also thank the open-source AI community for datasets, tools, and research support.

ABSTRACT

The rapid growth of social media platforms has significantly increased the spread of fake news and manipulated media content. Deepfake videos and AI-generated misinformation are becoming major challenges in the digital era. This project proposes an AI-Based Fake Reels & News Detector that uses Natural Language Processing, Deep Learning, Convolutional Neural Networks, and BERT-based classification models to detect fake textual news and manipulated reels automatically. The proposed system performs textual preprocessing, frame extraction, deepfake analysis, contextual understanding, and classification using advanced AI algorithms. The application provides REAL or FAKE predictions with confidence scores to improve digital trust and reduce misinformation spread. The system demonstrates high accuracy and efficient performance suitable for real-time deployment.

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CHAPTER 1 – INTRODUCTION

Section 1

The growth of Artificial Intelligence and social media has transformed the way information is shared across the internet. Millions of users consume online content every day through platforms such as Instagram, YouTube, Facebook, and X (Twitter). However, the increase in digital communication has also increased the spread of fake news, misinformation, and manipulated video content. Deepfake technology uses AI models to generate realistic fake videos and synthetic media. Fake news articles use misleading headlines, edited information, and emotional manipulation to influence users. Detecting such content manually is difficult because of the massive volume of online data. The AI-Based Fake Reels & News Detector is designed to solve this problem using Machine Learning, Deep Learning, and NLP techniques. Artificial Intelligence-based misinformation detection systems are becoming increasingly important due to the rapid growth of AI-generated content. The integration of Deep Learning and Natural Language Processing provides efficient solutions for analyzing textual and visual media content. The proposed framework improves contextual understanding, enhances scalability, and supports efficient prediction generation. Advanced transformer models such as BERT improve semantic understanding, while CNN-based image analysis improves manipulated frame detection. The implementation demonstrates how hybrid AI systems can solve real-world misinformation problems effectively.

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CHAPTER 2 – LITERATURE SURVEY

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Researchers across the world have proposed various techniques for fake news and deepfake detection. Traditional systems used keyword-based filtering and manual verification. Modern approaches use transformer models such as BERT and RoBERTa. Deepfake detection systems use CNN architectures, GAN detection methods, Autoencoders, and frame analysis techniques. Natural Language Processing techniques such as tokenization, stemming, lemmatization, and contextual embeddings improve textual analysis performance. Artificial Intelligence-based misinformation detection systems are becoming increasingly important due to the rapid growth of AI-generated content. The integration of Deep Learning and Natural Language Processing provides efficient solutions for analyzing textual and visual media content. The proposed framework improves contextual understanding, enhances scalability, and supports efficient prediction generation. Advanced transformer models such as BERT improve semantic understanding, while CNN-based image analysis improves manipulated frame detection. The implementation demonstrates how hybrid AI systems can solve real-world misinformation problems effectively.

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CHAPTER 3 – EXISTING SYSTEM

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CHAPTER 4 – PROPOSED SYSTEM

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The proposed system integrates NLP, CNN, OpenCV, TensorFlow, and BERT models to create a hybrid misinformation detection framework. The system workflow includes: 1. User Input 2. Preprocessing 3. Textual Analysis 4. Frame Extraction 5. Deepfake Detection 6. Confidence Score Generation 7. Result Visualization The application predicts whether the content is REAL or FAKE. Artificial Intelligence-based misinformation detection systems are becoming increasingly important due to the rapid growth of AI-generated content. The integration of Deep Learning and Natural Language Processing provides efficient solutions for analyzing textual and visual media content. The proposed framework improves contextual understanding, enhances scalability, and supports efficient prediction generation. Advanced transformer models such as BERT improve semantic understanding, while CNN-based image analysis improves manipulated frame detection. The implementation demonstrates how hybrid AI systems can solve real-world misinformation problems effectively.

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CHAPTER 5 – SYSTEM REQUIREMENTS

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Hardware Requirements: - Intel i5 Processor or higher - 8 GB RAM - 20 GB Storage - Internet Connectivity

Software Requirements: - Python - TensorFlow - OpenCV - Flask - VS Code - Jupyter Notebook Artificial

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CHAPTER 6 – SYSTEM DESIGN

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The architecture consists of multiple layers: - User Interface Layer - NLP Processing Layer - Video Analysis Layer - Prediction Engine - Database Layer The system follows modular architecture to improve scalability and maintainability. Artificial Intelligence-based misinformation detection systems are becoming increasingly important due to the rapid growth of AI-generated content. The integration of Deep Learning and Natural Language Processing provides efficient solutions for analyzing textual and visual media content. The proposed framework improves contextual understanding, enhances scalability, and supports efficient prediction generation. Advanced transformer models such as BERT improve semantic understanding, while CNN-based image analysis improves manipulated frame detection. The implementation demonstrates how hybrid AI systems can solve real-world misinformation problems effectively.

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CHAPTER 7 – SYSTEM IMPLEMENTATION

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CHAPTER 8 – TESTING

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The testing phase validates: - Fake news detection accuracy - Deepfake detection performance - Real-time response capability - UI responsiveness - Error handling Performance metrics include Accuracy, Precision, Recall, and F1-Score. Artificial Intelligence-based misinformation detection systems are becoming increasingly important due to the rapid growth of AI-generated content. The integration of Deep Learning and Natural Language Processing provides efficient solutions for analyzing textual and visual media content. The proposed framework improves contextual understanding, enhances scalability, and supports efficient prediction generation. Advanced transformer models such as BERT improve semantic understanding, while CNN-based image analysis improves manipulated frame detection. The implementation demonstrates how hybrid AI systems can solve real-world misinformation problems effectively.

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CHAPTER 9 – RESULTS AND DISCUSSION

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CHAPTER 10 – CONCLUSION AND FUTURE ENHANCEMENT

Section 1

The project successfully demonstrates how Artificial Intelligence can be used to reduce misinformation spread and improve digital trust. Future enhancements include: - Multilingual support - Real-time social media integration - Blockchain verification - Mobile application deployment Artificial Intelligence-based misinformation detection systems are becoming increasingly important due to the rapid growth of AI-generated content. The integration of Deep Learning and Natural Language Processing provides efficient solutions for analyzing textual and visual media content. The proposed framework improves contextual understanding, enhances scalability, and supports efficient prediction generation. Advanced transformer models such as BERT improve semantic understanding, while CNN-based image analysis improves manipulated frame detection. The implementation demonstrates how hybrid AI systems can solve real-world misinformation problems effectively.

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Section 4

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while CNN-based image analysis improves manipulated frame detection. The implementation demonstrates how hybrid AI systems can solve real-world misinformation problems effectively.

PERFORMANCE ANALYSIS

Metric	Value
Accuracy	94%
Precision	92%
Recall	91%
F1 Score	93%

APPENDIX

Python Fake News Prediction Example:

```
model.predict(news_input)
```

OpenCV Frame Extraction:

```
cv2.VideoCapture(video_path)
```

TensorFlow CNN Prediction:

```
cnn_model.predict(frame_data)
```

REFERENCES

1. Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers." 2. Vaswani et al., "Attention Is All You Need." 3. Ian Goodfellow et al., "Deep Learning." 4. TensorFlow Official Documentation. 5. OpenCV Official Documentation. 6. Kaggle Fake News Dataset. 7. FaceForensics++ Research Dataset. 8. IEEE Research Papers on Deepfake Detection. 9. NLP Research Publications. 10. CNN and Computer Vision Research Articles.